

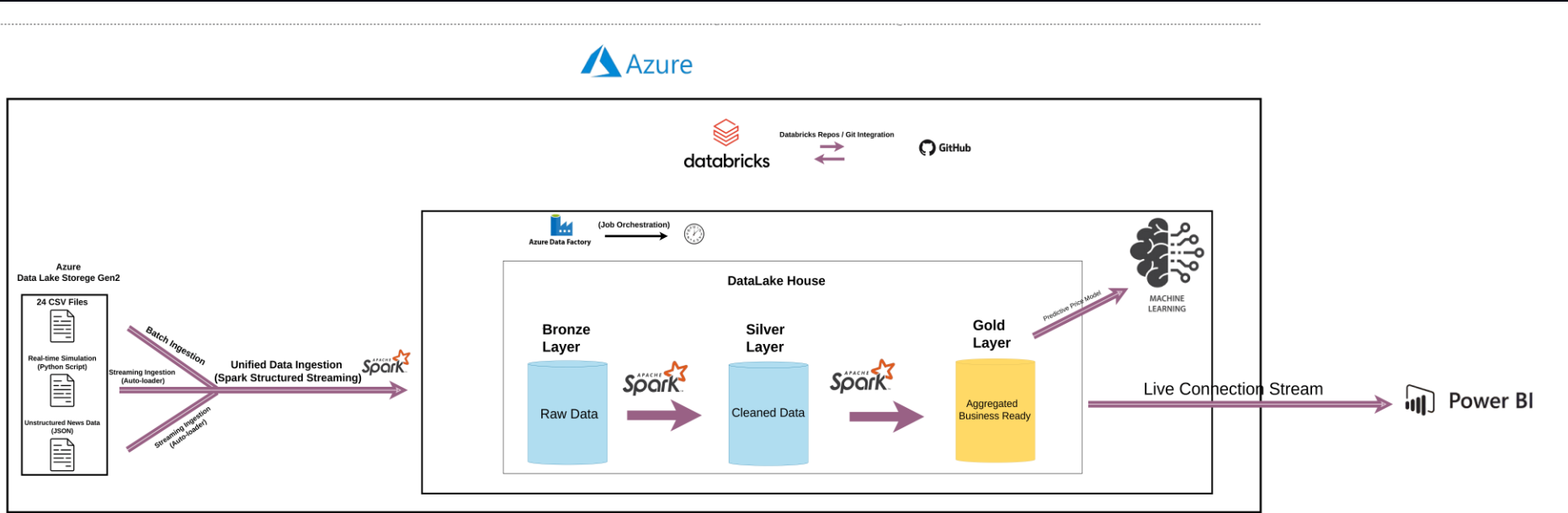
UNIFIED CRYPTO-INTELLIGENCE LAKEHOUSE LAKEHOUSE

Building Real-Time Insights & Predictive Analytics with Databricks and Apache Spark

01 // PROJECT DESCRIPTION

This project aims to build an **end-to-end data engineering pipeline** to analyze the cryptocurrency market (Bitcoin) in real-time. The system integrates multiple data streams, including **historical price data (Batch)**, **live price updates (Streaming)**, and **unstructured news data (JSON)**. By leveraging the **Medallion Architecture**, the project transforms raw, chaotic data into structured insights. A key highlight is the integration of **Natural Language Processing (NLP)** to perform sentiment analysis on market news, combined with **Machine Learning** to predict price trends based on both financial metrics and public sentiment.

02 // PROJECT ARCHITECTURE



06 // PREDICTIVE ANALYTICS & ML

PHASE_01

FEATURE ENG.

Transforming curated Gold-layer data into high-dimensional features for model training.

TECHNICAL

Moving Averages (MA), RSI, and Volatility metrics metrics.

SENTIMENT

Aggregated Sentiment Index from real-time news news feeds.

PHASE_02

MODEL MGMT.

Utilizing MLflow for a professional MLOps workflow and experiment tracking.

```
mlflow.start_run()
mlflow.log_param("alpha", 0.5)
mlflow.log_metric("rmse", 0.12)
mlflow.end_run()
```

Ensuring reproducibility through versioned models and hyperparameter logging.

PHASE_03

INFERENCE

Generating real-time predictive signals based on the fusion of market and sentiment data.

Predicting next-hour price trends with confidence intervals.

BUY

HOLD

SELL

 Model updated every 15 minutes.

03 // MEDALLION ARCHITECTURE

DATA_TIER_01



BRONZE

The "Source of Truth" layer. Data is stored in its original format to allow for full lineage and reprocessing.



Raw Imutable History



Ingestion Metadata

DATA_TIER_02



SILVER

The "Engine Room." Data is cleaned, validated, and and enriched with NLP sentiment scores for analysis. analysis.



Deduplication & Cleaning



NLP Sentiment Scoring

DATA_TIER_03



GOLD

The "Analytics Layer." Curated, aggregated datasets datasets optimized for Power BI and Machine Learning. Learning.



Stream-Batch Joins



Business-Ready Views

03 // CORE TECHNOLOGY STACK

CLOUD PLATFORM



DATABRICKS

Unified analytics platform for data processing, processing, machine learning, and collaborative engineering.

DATA ENGINE

APACHE SPARK

Distributed engine for high-speed batch and and stream processing using PySpark and SQL and SQL.

STORAGE LAYER



DELTA LAKE

Providing ACID transactions and scalable metadata handling for the unified lakehouse. lakehouse.

STREAMING



SPARK STREAMING

Real-time data ingestion using the Auto-loader loader feature for low-latency updates.

AI & NLP



MLFLOW / VADER

Managing ML experiments and performing performing sentiment analysis on unstructured unstructured news data.

VERSION CONTROL



GITHUB

CI/CD integration and code repository management for robust engineering workflows.

ORCHESTRATION



WORKFLOWS

Automating pipeline jobs to ensure 24/7 data data flow and operational reliability with Azure Azure Data Factory.

VISUALIZATION



POWER BI

Live-connected dashboards for real-time market monitoring and stakeholder insights. insights.

08 // ROBUST DATA FOUNDATION



PRIMARY DATA SOURCE

KAGGLE DATASET

Cryptocurrency Historical Prices by Sudalairajkumar

DATA FIDELITY

High-resolution historical price data (Open, High, Low, Close, Volume) Close, Volume) for BTC, ETH, and other major assets.



kaggle.com/datasets/sudalairajkumar/cryptocurrencypricehistory



CONTINUOUS INGESTION



DATABRICKS AUTO-LOADER

Efficiently processes new files as they arrive in the landing zone without manual intervention.



SCHEMA EVOLUTION

Automatically handles changes in data structure to ensure pipeline stability and reliability.



DATA INTEGRITY

Ensuring high-fidelity data for robust model training, validation, and real-time simulation.

09 // TEAM & FUTURE OUTLOOK

PROJECT TEAM



Youssef Hamed Abdelmonim
Abdelmonim Ahmed

TEAM LEADER



Seif Mohamed Fathi Abdelaziz
Abdelaziz



Yehia Yasser Yahya Salama



Nada Mahmoud Hammad
Ibrahim



Nourhan Ahmed Gaber
Sharawy

FUTURE HORIZON



MULTI-ASSET SUPPORT

Expanding the lakehouse architecture to ingest and analyze data for Ethereum, Solana, Ethereum, Solana, and other major digital assets.



ADVANCED PREDICTION

Implementing deep learning models (LSTMs/Transformers) for higher-accuracy price accuracy price forecasting and risk assessment.



GLOBAL INTELLIGENCE

Integrating global macroeconomic indicators and regulatory news for a truly holistic holistic crypto-intelligence platform.